Your Cancer Summary Report

Anonymous Patient (PAT001)

Stage IV High-Grade Serous Ovarian Cancer

MCP Precision Oncology Research Platform

Report Date: March 10, 2026

⚠️ **DRAFT - FOR CLINICIAN REVIEW**

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Ovarian Cancer - High-Grade Serous Carcinoma (HGSOC)

Stage: Stage IV | **Grade:** High Grade

You have an advanced form of ovarian cancer called high-grade serous carcinoma. This type has spread beyond the ovaries. While this is a serious diagnosis, your tumor carries important genetic markers that open the door to several targeted therapies — including some newer approaches identified by our precision medicine platform.

✓ Positive Factors

- BRCA1 pathogenic mutation makes you eligible for PARP inhibitor maintenance therapy (olaparib) — a targeted drug with strong evidence in HGSOC
- HRD score of 72 (well above the ≥42 eligibility threshold) further confirms PARP inhibitor benefit
- Strong neoantigen identified (RMPEAAPPV, IC50 = 7.8 nM) — your immune system has a target it may be able to attack
- Antigen presentation pathway score 0.76/1.0 indicates intact machinery for immune recognition
- Spatial analysis identified CD8+ T cells concentrated in the stroma — immune cells are present and may be recruited into the tumor with the right therapy

- Tumor core shows lower CD8+ T cell density than surrounding tissue, suggesting immune cells are being kept out (immune exclusion pattern)
- TMB of 4.2 mut/Mb falls below the ≥10 mut/Mb threshold for standard pembrolizumab eligibility
- CAF (cancer-associated fibroblast) content at 18.2% may contribute to treatment resistance

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA1

Pathogenic

Pathogenic germline mutation

You carry an inherited change in the BRCA1 gene. This gene normally helps repair DNA damage. When it does not work properly, cancer cells become dependent on a backup repair system — and drugs called PARP inhibitors can shut down that backup system, killing cancer cells while sparing healthy ones.

Treatment Option: FDA-approved PARP inhibitors: olaparib (Lynparza), niraparib (Zejula), rucaparib (Rubraca) for maintenance therapy

TP53

Pathogenic

R175H missense mutation

This is the most common mutation in ovarian cancer. The TP53 gene normally acts as a 'guardian' that tells damaged cells to stop growing. When it is mutated, cells can grow unchecked. Importantly, the specific R175H change in TP53 creates a unique protein fragment (peptide) that the immune system can recognize — this is the basis for a potential neoantigen vaccine.

Treatment Option: Neoantigen peptide RMPEAAPPV identified as strong HLA-A*02:01 binder (IC50 = 7.8 nM); clinical trials available

AKT1

Likely Pathogenic

Pathway activation (multi-omics)

Our multi-omics analysis found that the AKT1/MTOR signaling pathway — which controls cell growth — is overactive in your tumor. This is not a single mutation but a pattern seen across multiple data types. Drugs that block this pathway (PI3K/AKT inhibitors such as alpelisib) may be worth exploring in clinical trials.

Treatment Option: Alpelisib (PI3K inhibitor) identified as top upstream drug candidate (Z-score = 9.23); Clinical Trial Only at this time for HGSOC

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment
Analysis of 300 spatial transcriptomics spots across 4 tissue regions (tumor core, stroma, immune zone, necrotic) revealed a tumor-cell-dominant microenvironment with significant stromal and immune components. Cell-type deconvolution: tumor cells 56 spots (19%), endothelial cells 44 spots (15%), macrophages 43 spots (14%), fibroblasts/CAFs 41 spots (14%), CD8+ T cells 30 spots (10%). Spatial autocorrelation (Moran's I ≈ −0.003 for all major markers) indicates a spatially uniform, diffuse expression distribution consistent with infiltrating growth pattern.

Immune Status: ● Mixed immune activity - Variable immune response across the tumor

Our spatial mapping of your tumor tissue found immune cells (CD8+ T cells) present in the tissue around your tumor but blocked from entering the tumor itself. Think of it like an army that is stationed outside the castle but cannot get through the walls. This pattern — called immune exclusion — means that if we can find a way to open the gates (for example, with checkpoint inhibitor drugs or CAF-disrupting agents), those immune cells may be able to attack the cancer. We also found that your tumor has good blood vessel activity and a layer of supporting cells called fibroblasts, both of which are targets for existing and experimental therapies.

Key Patterns Observed

- CD8A immune exclusion: 493 TPM in stroma vs. 89 TPM in tumor core (5.5× gradient) — T cells are present but kept out of the tumor
- VEGFA spatially clustered (Moran's I = 0.063) within t

- Macrophage infiltration: 43 spots (14%) with TAM signature — immune modulation potential
- Near-uniform spatial expression (Moran's I ≈ −0.003) consistent with diffuse infiltrating HGSOC growth pattern

What the Microscope Shows

Microscopy of your tumor tissue confirms that approximately one in three cancer cells is actively dividing at any given time (a measure called the Ki67 index). The TP53 protein staining pattern matches your R175H genetic mutation. Immune cells called CD8+ T cells were visible in the tissue but concentrated away from the main tumor — consistent with the immune exclusion pattern identified in our spatial analysis.

Key Observations

- Nuclear segmentation of DAPI-stained tissue identified distinct cell populations
- Ki67 immunofluorescence: 34% proliferating cells (Ki67-positive), 66% quiescent — moderate-to-high proliferative index consistent with HGSOC
- TP53 immunofluorescence: positive staining in tumor cell nuclei consistent with R175H missense mutation
- CD8A immunofluorescence: positive T cell infiltration present predominantly in stromal regions

Treatment Options

Based on your test results, your care team may consider these treatments:

Carboplatin + Paclitaxel (Standard Chemotherapy)

chemotherapy

NCCN Category 1

Expected Response: 70-80%

This is the standard first-line treatment for ovarian cancer. Two chemotherapy drugs are given by IV, usually every 3 weeks for 6 cycles. They work by damaging the DNA of rapidly dividing cancer cells.

Why this may help you: Standard NCCN Category 1 first-line regimen for Stage IV HGSOC. Your BRCA1 mutation status makes platinum-based chemotherapy particularly effective due to shared DNA repair pathway dependencies.

Common side effects: Hair loss (usually temporary), Nausea and vomiting, Fatigue, Increased infection risk, Numbness or tingling in hands and feet (neuropathy)

Bevacizumab (Avastin) Maintenance

targeted

NCCN Category 1

Expected Response: Extended progression-free survival by ~4 months

Bevacizumab is a targeted therapy that blocks a protein called VEGF, which tumors need to build new blood vessels. By cutting off the blood supply, it can slow tumor growth. It is given by IV every 3 weeks, often alongside and after chemotherapy.

Why this may help you: Recommended for high-risk Stage IV HGSOC. Spatial analysis showed VEGFA clustering in your tumor (Moran's I = 0.062), and your tumor microenvironment includes significant angiogenic activity. NCCN Category 1 recommendation for Stage IV disease.

Common side effects: High blood pressure, Fatigue, Increased bleeding risk, Delayed wound healing, Protein in urine

Olaparib (Lynparza) Maintenance

targeted

FDA Approved

Expected Response: 70% reduction in progression risk vs. placebo (SOLO-1 trial)

Olaparib is a PARP inhibitor — a pill taken twice daily that targets cancer cells with BRCA mutations. Because your cancer cells cannot properly repair their DNA (due to BRCA1 mutation), olaparib causes them to accumulate damage until they die. Healthy cells with working BRCA1 can survive this treatment.

Why this may help you: Your BRCA1 pathogenic mutation and HRD score of 72 (threshold ≥42) make this an FDA-approved standard-of-care recommendation with strong evidence. The SOLO-1 trial showed 70% reduction in progression risk for BRCA-mutated HGSOC.

Common side effects: Nausea (usually manageable), Fatigue, Anemia (low red blood cells), Low appetite, Rarely: a blood cancer called MDS (less than 1%)

Neoantigen Peptide Vaccine (RMPEAAPPV — Investigational)

immunotherapy

Clinical Trial Only

Your TP53 R175H mutation creates a unique protein fragment (called a neoantigen) that your immune system can potentially recognize as foreign. A personalized vaccine could be made from this fragment to train your immune system to attack cancer cells carrying this mutation. This is an emerging approach being tested in clinical trials.

Why this may help you: Our precision platform identified the peptide RMPEAAPPV as a strong binder to your HLA-A*02:01 immune type (IC50 = 7.8 nM, well below the 50 nM strong-binder threshold). Your antigen presentation score of 0.76/1.0 indicates intact machinery to present this peptide to immune cells. This is a pipeline-identified novel path not available through standard workup.

Common side effects: Injection site redness or swelling, Mild fever or flu-like symptoms, Fatigue

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Checkpoint Inhibitor (Spatial CD8A Strategy — Investigational)

immunotherapy

Clinical Trial Only

Your immune cells (CD8+ T cells) are present in the tissue surrounding your tumor but are being kept out of the tumor itself — a pattern called immune exclusion. Certain drugs can help break down this barrier and let immune cells into the tumor, where they can fight cancer. Combining checkpoint inhibitors with stromal-disrupting agents is being studied in clinical trials.

Why this may help you: Spatial transcriptomics showed CD8A expression at 493 TPM in stroma vs 89 TPM in tumor core — a 5.5× gradient indicating immune exclusion. This makes your tumor a candidate for checkpoint therapy strategies aimed at overcoming immune exclusion. This finding is only possible with spatial analysis and is a novel path from our platform.

Common side effects: Immune-related side effects (rash, colitis, thyroid changes), Fatigue, Infusion reactions

NNMT Inhibitor / CAF-Targeting (Investigational)

targeted

Clinical Trial Only

Your tumor has a significant population of cancer-associated fibroblasts (CAFs) — supporting cells that help the cancer grow and hide from treatments. Drugs that target these fibroblasts, including NNMT inhibitors, are in early clinical trials and may help improve responses to other treatments by changing the tumor's environment.

Why this may help you: Spatial deconvolution identified 18.2% fibroblast/CAF content, placing your tumor in the CAF-high stratum. Research from GSE292142 (2025 HGSOC NNMT dataset) shows NNMT is upregulated in these tumors. This is a novel path identified by our platform not accessible through standard workup.

Common side effects: Under investigation — side effect profile still being characterized in early trials

Alpelisib (PI3K Inhibitor — Investigational for HGSOC)

targeted

Clinical Trial Only

Alpelisib blocks an overactive cellular growth signal called the PI3K/AKT pathway. Our multi-omics analysis found this pathway highly activated in your tumor (ranked #1 upstream drug candidate, Z-score = 9.23). While alpelisib is FDA-approved for breast cancer with PIK3CA mutations, it is currently investigational for ovarian cancer.

Why this may help you: Multi-omics upstream regulator analysis identified AKT1/MTOR pathway activation with alpelisib as the top predicted drug response (Z-score = 9.23). This is not detectable on standard clinical workup and requires integrated RNA + protein data analysis — a novel path from our precision platform.

Common side effects: High blood sugar (hyperglycemia) — requires monitoring, Nausea, Diarrhea, Rash, Fatigue

Clinical Trials

You may be eligible for these research studies offering new treatments:

Olaparib Plus Pembrolizumab in BRCA-Mutated Ovarian Cancer
Phase: 2 | **Status:** RECRUITING

Trial ID: NCT04644068

This trial tests a combination of your recommended PARP inhibitor (olaparib) with an immunotherapy drug. Given your intact antigen presentation pathway (score 0.76) and BRCA1 mutation, you may benefit from this combination. Your care team can check whether your specific staging and prior treatment history qualify.

Nearest Location: Multiple US sites

Contact: ClinicalTrials.gov NCT04644068

Personalized Neoantigen Peptide Vaccine in Solid Tumors with TP53 Mutations
Phase: 1 | **Status:** RECRUITING

Trial ID: NCT05100355

This early-phase trial tests personalized vaccines made from TP53 mutations like your R175H variant. Your neoantigen RMPEAAPPV was identified as a strong binder to your immune type (HLA-A*02:01), making you a good candidate. This is an investigational approach and requires enrollment at a participating center.

Nearest Location: Select academic medical centers

Contact: ClinicalTrials.gov NCT05100355

Alpelisib Plus Carboplatin in PI3K-Activated Gynecologic Cancers
Phase: 2 | **Status:** RECRUITING

Trial ID: NCT05308056

Your multi-omics analysis identified PI3K/AKT pathway activation as a top target (alpelisib Z-score = 9.23). This trial tests alpelisib combined with chemotherapy specifically in ovarian and other gynecologic cancers with evidence of PI3K pathway activation — which your tumor shows.

Nearest Location: NCI-designated cancer centers

Contact: ClinicalTrials.gov NCT05308056

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 3 months	CA-125 blood test CA-125 is a protein made by ovarian cancer cells. Tracking it over time shows whether treatment is working or if cancer is returning.
Every 6 months, or sooner if symptoms arise	CT scan (chest, abdomen, pelvis) Imaging confirms what the CA-125 blood test suggests and shows exactly where cancer is or is not present.
Before each treatment cycle, then monthly on maintenance	Complete blood count (CBC) Monitors for anemia and infection risk, especially important during olaparib maintenance therapy.
Every 3 months	Kidney and liver function panel Checks that your kidneys and liver are processing treatment drugs safely.
Weekly for first month of bevacizumab, then each visit	Blood pressure check Bevacizumab can raise blood pressure, which needs prompt management.
Once (strongly recommended), with follow-up as needed	Genetic counseling session Your BRCA1 mutation is hereditary. A genetic counselor can explain what this means for you and your blood relatives, and help arrange family testing.

⚠️ When to Call Your Doctor

Contact your care team right away if you experience:

- New or worsening numbness or tingling in hands or feet
- Fever over 38°C (100.4°F) — may signal infection during low white blood cell periods
- Unusual bleeding or bruising
- Significant fatigue that prevents normal activities
- Nausea or vomiting that prevents eating or drinking

Questions? Your oncology nurse navigator or care team — available during business hours. For urgent symptoms, proceed to the emergency department and inform them you are receiving cancer treatment.

This AI-generated summary was produced by a precision oncology research platform and must be reviewed and approved by your healthcare team before any treatment decisions. It is not a substitute for professional medical advice. All novel findings (neoantigen vaccine, spatial immune exclusion, NNMT inhibition, alpelisib) require prospective clinical validation.

For Your Family

Your BRCA1 mutation appears to be a germline (inherited) mutation, meaning it was present in all of your cells from birth and was passed down through your family. Each of your first-degree relatives (parents, siblings, children) has a 50% chance of also carrying this mutation. Carrying BRCA1 significantly increases the lifetime risk of ovarian cancer (~44%) and breast cancer (~72%). Relatives who test positive can take preventive steps, including increased screening or risk-reducing surgery.

Recommended Steps for Family Members

- Share this finding with your first-degree relatives (parents, siblings, children) and encourage them to speak with their doctors
- Consider a genetic counseling appointment for yourself to fully understand the implications and options
- Female relatives who test positive may benefit from enhanced breast/ovarian screening starting at age 25-30
- Male relatives who test positive have increased risk of breast cancer and prostate cancer

> **Genetic Counseling Recommended:** A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:

📣 **FORCE (Facing Our Risk of Cancer Empowered)**

National nonprofit supporting people with BRCA mutations and hereditary cancer. Provides peer support, educational resources, and help navigating treatment decisions.

📞 1-866-288-7475 | 🌐 https://www.facingourrisk.org

👥 **Ovarian Cancer Research Alliance (OCRA)**

Ovarian cancer-specific patient support, education about clinical trials, and connection to patient communities and peer navigators.

📞 1-866-399-6262 | 🌐 https://ocrahope.org

🧬 **Genetic Counseling Referral**

Board-certified genetic counselors can help you and your family understand the BRCA1 mutation, testing options for relatives, and risk management strategies. Ask your oncologist for a referral, or find one at the NSGC website.

| 🌐 https://www.nsgc.org/findageneticcounselor

💰 **NeedyMeds — Cancer Treatment Cost Assistance**

Connects patients with manufacturer patient assistance programs, co-pay cards, and other financial support for cancer treatments including olaparib and bevacizumab.

📞 1-800-503-6897 | 🌐 https://www.needymeds.org

💚 **Cancer Care Oncology Social Work**

Free professional counseling, support groups, and educational workshops for people with cancer and their families. Oncology social

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Data Sources: mcp-mockepic (patient records), mcp-fgbio (genomic analysis), mcp-genomic-results (somatic variants + HRD score), mcp-spatialtools (spatial transcriptomics, 300 spots), mcp-multiomics (RNA + protein integration), mcp-neoantigen (MHC-I binding prediction), mcp-cell-classify (IF cell segmentation), mcp-opentargets (target-disease associations), mcp-mocktcga (TCGA cohort comparison), mcp-clinical-trials (trial matching)
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